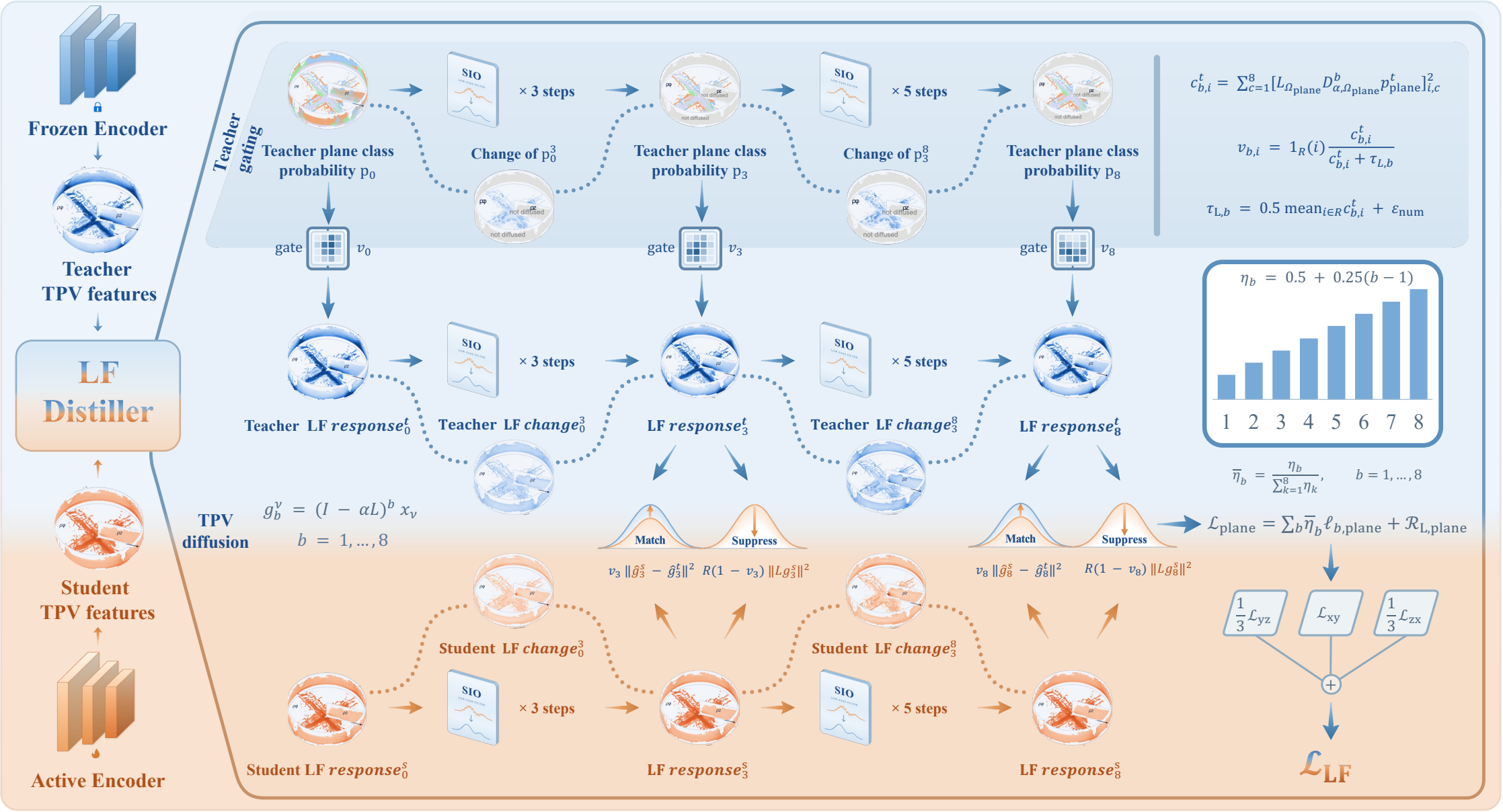

Frozen Encoder
Teacher TPV features
LF Distiller
Student TPV features
Active Encoder
Teacher gating
Teacher plane class probability p_0
Change of p_0^3
Teacher plane class probability p_3
Change of p_3^8
Teacher plane class probability p_8
× 3 steps
× 5 steps
gate
v_0
v_3
v_8
Teacher LF response$_0^t$
Teacher LF change$_0^3$
LF response$_3^t$
Teacher LF change$_3^8$
LF response$_8^t$
TPV diffusion
$g_b^v = (I - \alpha L)^b x_v$
$b = 1, \dots, 8$
Match
Suppress
$v_3 \|\hat{g}_3^s - \hat{g}_3^t\|^2$ $R(1 - v_3) \|Lg_3^s\|^2$
$v_8 \|\hat{g}_8^s - \hat{g}_8^t\|^2$ $R(1 - v_8) \|Lg_8^s\|^2$
Student LF change$_0^3$
Student LF change$_3^8$
Student LF response$_0^s$
LF response$_3^s$
LF response$_8^s$
$c_{b,i}^t = \sum_{c=1}^{8} [L_{\Omega_{\text{plane}}} D_{\alpha,\Omega_{\text{plane}}}^b p_{\text{plane}}^t]_{i,c}^2$
$v_{b,i} = 1_R(i) \frac{c_{b,i}^t}{c_{b,i}^t + \tau_{L,b}}$
$\tau_{L,b} = 0.5 \, \text{mean}_{i \in R} c_{b,i}^t + \varepsilon_{\text{num}}$
$\eta_b = 0.5 + 0.25(b - 1)$
1 2 3 4 5 6 7 8
$\overline{\eta}_b = \frac{\eta_b}{\sum_{k=1}^{8} \eta_k}$, $b = 1, \dots, 8$
$\mathcal{L}_{\text{plane}} = \sum_b \overline{\eta}_b \ell_{b,\text{plane}} + \mathcal{R}_{\text{L,plane}}$
$\frac{1}{3}\mathcal{L}_{yz}$ $\mathcal{L}_{xy}$ $\frac{1}{3}\mathcal{L}_{zx}$
$\mathcal{L}_{\text{LF}}$